DATA ENGINEERING

ETL Data Pipelines Visual Guide

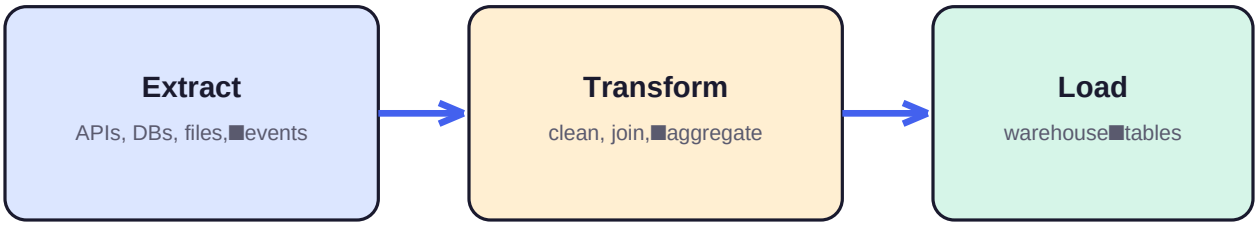
Extract, Transform, Load —
how raw data becomes trusted analytics.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

The Pipeline: E → T → L

Three stages, one direction: raw → clean → queryable.



Raw: messy, nested, duplicated. → Clean: typed, deduped, conformed. → Modeled: star schema, ready for BI.

ELT variant: Load raw first, Transform inside the warehouse (dbt). Better for huge data + cheap compute.

STEP 2/6

Extract: Getting Data Out

Every source has a personality.

Databases (CDC)

Change-data-capture streams row changes (Debezium) — no polling.

APIs

REST/GraphQL pulls; respect rate limits, paginate, backfill.

Files

CSV/Parquet drops in S3; watch for schema drift.

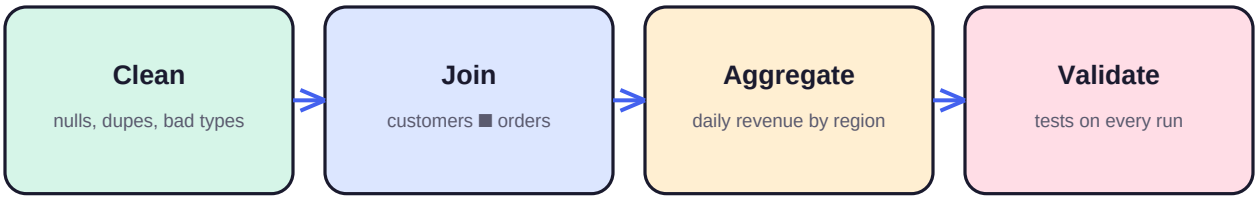
Events

Clickstreams, IoT — high volume, append-only.

STEP 3/6

Transform: The Real Work

Where data quality is made or broken.

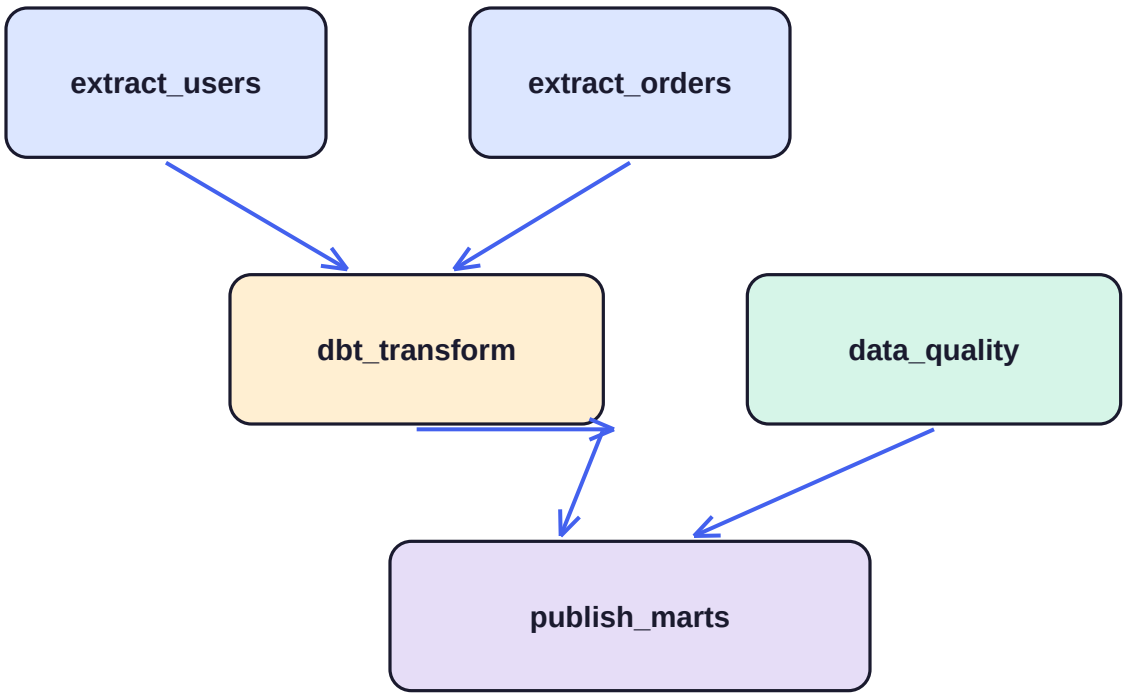


- Idempotent transforms: re-running produces the same result (no double-counting).
- dbt: SQL models with tests, docs, and lineage — versioned in git.
- Slowly-changing dimensions: track history (Type 2) vs overwrite (Type 1).

STEP 4/6

Orchestration: DAGs Run the Show

Dependencies as a directed acyclic graph.



Airflow / Dagster / Prefect: schedule, retry, alert, backfill. A failed task retries; downstream waits.

STEP 5/6

Data Quality Tests

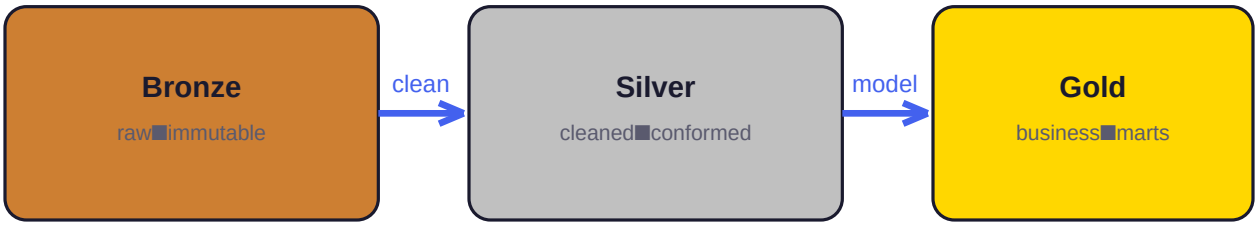
Trust is earned per run.

- **Freshness**
  Did today's data arrive? (SLA breach → alert)
- **Volume**
  Row counts within expected bands (no silent drops)
- **Schema**
  Columns and types match the contract
- **Uniqueness**
  Keys are actually unique; no fan-out joins

STEP 6/6

Medallion Architecture

Bronze → Silver → Gold: progressive refinement.



Bronze: keep everything raw — you can always re-derive. Never delete source truth.

Gold: curated, documented, SLAd — the only layer analysts touch.

STEP ✓/✓

Key Takeaways

ETL Pipelines — Remember

- Extract from DBs (CDC), APIs, files, events — each with its own quirks.
- Transform idempotently; dbt brings software practices to SQL.
- Orchestrate with DAGs: dependencies, retries, backfills, alerts.
- Test every run: freshness, volume, schema, uniqueness.
- Medallion: bronze (raw) → silver (clean) → gold (business-ready).

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev